Problem Statement

Wisconsin Breast Cancer Dataset:

Build an automated machine learning model that accurately predicts whether a tumor is Benign (non-cancerous) or Malignant (cancerous).

Dataset Description

1. The Input Features (X):

The dataset is provided with highly detailed, multi-dimensional numerical features collected from the microscopic images of cell nuclei during a biopsy. These include attributes like:

- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- Marginal Adhesion
- Single Epithelial Cell Size
- Bare Nuclei
- Bland Chromatin
- Normal Nucleoli
- Mitoses

2. The Target Outputs (y)

The model must map those numeric features into one of two strict target classes:

- 0 (or 2 in the raw data): Benign (safe mass, minimal intervention needed).
- 1 (or 4 in the raw data): Malignant (cancerous mass, immediate clinical treatment required).

Github Repository Link

https://github.com/pr701368/Classifiers

Models used

Metrics obtained after running belwo models on Wisconsin Breast Cancer Dataset

Model	Accuracy	Precision	Recall (Sensitivity)	F1-Score	Matthews Corr Coeff (MCC)	AUC
Logistic Regression	0.8946	0.822	0.9151	0.8661	0.7826	0.9219
Decision Tree Classifier	1	1	1	1	1	1
K-Nearest Neighbor Classifier	0.9473	0.955	0.9009	0.9272	0.8868	0.9917
Naive Bayes Classifier	0.9947	0.986	1	0.993	0.9888	1
Support Vector Machine Classifier	0.7504	0.6	0.9906	0.7473	0.5946	0.7992

Observations

Model	Observations
Logistic Regression	a solid linear baseline, hitting an Accuracy of 0.8946. While it performs reasonably well across the board, its slightly lower Precision (0.8220) and F1-Score (0.8661) mean it occasionally struggles to resolve non-linear overlap near the core decision boundaries.
Decision Tree Classifier	a perfect score on a training/test set combination usually indicates a massive Overfitting problem. Because the custom decision tree evaluates splits until data points are perfectly separated (without structural pruning), it has likely memorized the exact noise in the training set rather than learning generalizable patterns
K-Nearest Neighbor Classifier	overall balance, achieving an Accuracy of 0.9473. It displays excellent diagnostic reliability with a high Precision of 0.9550 and a balanced F1-Score of 0.9272, proving that clusters of identical cells align closely in multi-dimensional space. But it is cost expensive
Naive Bayes Classifier	best reliable performer, scoring near-perfect metrics with an Accuracy of 0.9947, Precision of 0.9860, and a perfect Recall of 1.0. This indicates the probability distributions of the breast cancer features map incredibly well onto Gaussian assumptions. It catches every single malignant case (Recall = 1.0) while generating almost zero false alarms
Support Vector Machine Classifier	the weakest model overall, yielding a low Accuracy of 0.7504 and Precision of 0.6000. It over-predicts malignancy to stay safe. This yields a strong Recall of 0.9906 but floods the user with stressful false positives

Overall Winner of dataset: Naive Bayes Classifier

Streamlit app features:

Streamlit deployed app url/link:

https://classifiers-xvf54siefuhvdnx35u8uwr.streamlit.app/

##

a. Dataset upload option: By default, when page loads, all or the selected models run on Wisconsin Breast Cancer Dataset. Also, a upload button available to upload datasets and apply predictions on it.

b. Model Select Option: A dropdown available to select models to run on dataset. By default, all are selected. We can remove one or more models and apply the remaining models on the dataset.

c. Display of Evaluation Metrics: Shows metrics of models applied on dataset in th ewebpage

d. Confusion matrix or classification report: Contains a dropdown to select model. On selecting a model, a report displayed with cases predicted positive and negative

Machine Learning Classifier Benchmark Application

An interactive benchmarking suite that loads ****5 saved classifier models**** from disk and evaluates them across ****6 core classification metrics**** using a live Streamlit dashboard interface.

📁 Project Structure

```text
project-folder/
├── app.py              # Streamlit dashboard interface code
├── requirements.txt    # Core environment library requirements
├── README.md           # Setup and project guide documentation
├── test_data.csv       # Evaluation hold-out dataset matrix
└── model/              # Directory containing serialized model pipelines
    ├── logistic_regression.joblib
    ├── decision_tree_classifier.joblib
    ├── k-nearest_neighbor_classifier.joblib
    ├── naive_bayes_gaussian.joblib
    └── support_vector_machine.joblib
```

⚙️ Setup and Installation

Follow these steps to run the pipeline on your local system environment:

Step 1: Clone or Navigate to the Folder

Open your terminal application and route into your active project root directory folder:

```bash
cd path/to/your/project-folder
```

Step 2: Install Libraries

Install the designated runtime dependencies listed in the requirements file using `pip`:

```bash
pip install -r requirements.txt
```

🚀 Execution Guide

Launching the Dashboard App

Start the interactive local hosting server by executing the dashboard command:

```bash
streamlit run app.py
```

Once running, the terminal will provide a local web URL (typically `http://localhost:8501`). Open this address inside your browser to access the complete metric comparison grid and charts.

📊 Implemented System Features

Evaluated Model Paradigms

1. **Logistic Regression** (Linear parametric probability categorization)
2. **Decision Tree Classifier** (Axis-aligned Gini structural branches)
3. **K-Nearest Neighbor Classifier** (Instance distance proximity mapping)
4. **Gaussian Naive Bayes** (Independent feature Bayes probability curves)

5. **Support Vector Machine** (Maximum-margin structural separating hyperplanes)

Evaluated Mathematical Performance Parameters

* **Accuracy**

* **AUC Score** (Computed via non-library Wilcoxon rank-sum ranking evaluation logic)

* **Precision**

* **Recall**

* **F1-Score** (Harmonic evaluation mean parameter metric)

* **Matthews Correlation Coefficient (MCC Score)**